1. **Linear Regression (Simple: 1 Independent Variable)**

Scenario: Predict test score (Y) from study hours (X).

Data:

Study Hours (X)	**Test Score (Y)**
1	60
2	75
3	85
4	95

Model: $Y = \beta_0 + \beta_1 X$

- Slope (β_1):

$$\beta_1 = \frac{n\sum(XY) - \sum X \sum Y}{n\sum(X^2) - (\sum X)^2} = \frac{4(1 \cdot 60 + 2 \cdot 75 + 3 \cdot 85 + 4 \cdot 95) - (10)(315)}{4(1 + 4 + 9 + 16) - 10^2} = 12.5$$

- Intercept (β_0):

$$\beta_0 = \bar{Y} - \beta_1 \bar{X} = 78.75 - 12.5 \times 2.5 = 47.5$$

Final Model: $TestScore = 47.5 + 12.5X$

Prediction for $X = 2.5$: $47.5 + 12.5 \times 2.5 = 78.75$.

2. **Logistic Regression (Binary Classification)**

Scenario: Predict heart disease ($Y = 1$: Yes, $Y = 0$: No) from exercise hours (X).

Data:

Exercise Hours (X)	**Heart Disease (Y)**
1	1
3	1
5	0

7	0

Model: $\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 X$

- Coefficients: $\beta_0 = 2, \beta_1 =- 0.8$ (simplified)

Probability: $P(Y = 1) = \frac{1}{1+e^{-(\beta_0+\beta_1 X)}}$

Prediction for $X = 4$:

$$P(Y = 1) = \frac{1}{1 + e^{-(2-0.8\times4)}} = \frac{1}{1 + e^{0.2}} \approx 0.45$$

→ Classified as "No Disease" ($0.45 < 0.5$) .

3. **Polynomial Regression (Quadratic)**

Scenario: Predict ice cream sales (Y, $)fromtemperature($X$, °C).

Data:

Temperature (X)	**Sales (Y)**
15	200
25	450
35	380

Model: $Y = \beta_0 + \beta_1 X + \beta_2 X^2$

- Coefficients: $\beta_0 =- 1000, \beta_1 = 80, \beta_2 =- 1.2$

Final Model: $Sales =- 1000 + 80X - 1.2X^2$

Prediction for $X = 30$: $-1000 + 80 \times 30 - 1.2 \times 30^2 = 320$ → $320 .

4. **Multiple Linear Regression (2+ Variables)**

Scenario: Predict house price (Y, $)fromsquarefootage($X_1$)andbedrooms($X_2$).

Data:

Sq Ft (X_1)	**Bedrooms (X_2)**	**Price (Y)**
1000	2	200,000

1500	3	280,000
2000	4	350,000

Model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

- Coefficients: $\beta_0 = 50,000$, $\beta_1 = 120$, $\beta_2 = 15,000$

Final Model: $Price = 50,000 + 120X_1 + 15,000X_2$
Prediction for $X_1 = 1800$, $X_2 = 3$: $50,000 + 120 \times 1800 + 15,000 \times 3 = 311,000 \rightarrow$ \$311,000 .

5. **Ridge Regression (L2 Penalty)**

Scenario: House price prediction with overfitting risk. Adds penalty $\lambda \sum \beta_i^2$.

- Original coefficients: $\beta_0 = 50,000$, $\beta_1 = 120$, $\beta_2 = 15,000$
- With $\lambda = 10$: Shrunken to $\beta_0 = 49,800$, $\beta_1 = 118$, $\beta_2 = 14,800$

Key: Reduces coefficients without setting to zero.
Prediction: $49,800 + 118 \times 1800 + 14,800 \times 3 = 306,600 \rightarrow$ \$306,600 [1][6].

6. **Lasso Regression (L1 Penalty)**

Scenario: Same as Ridge but uses L1 penalty ($\lambda \sum |\beta_i|$) for feature selection.

- Original coefficients: $\beta_0 = 50,000$, $\beta_1 = 120$, $\beta_2 = 15,000$
- With $\lambda = 10$: May set weaker coefficients to zero (e.g., $\beta_2 = 0$ if insignificant).

Key: Performs feature selection by eliminating variables .

Key Citations from Evidence:

Simple/Multiple/Ridge/Lasso Regression: Coefficients derived from optimization techniques; Ridge/Lasso use penalties to handle multicollinearity/overfitting.
Logistic Regression: Uses sigmoid function for probability estimation; classification via threshold (e.g., 0.5) .
Polynomial Regression: Captures non-linear relationships via higher-order terms (e.g., X^2).